



ARTIFICIAL INTELLIGENCE

iour without human supervision.

Automotive. AI-assisted automotive technology has been employed in driverless cars. Recently, IBM team has developed an IoT for automotive—a program to eliminate driver errors through connectivity. Since many accidents are caused by human errors, researchers are trying ways to minimise human errors through AI algorithms. In the future, a car is likely to have well-packaged complex computer programs. Automated vehicles use AI, sensors and global positioning system coordinates to drive themselves without a human operator.

Healthcare. AI and the IoT find applications in healthcare facilities and maintaining patient health records, medical equipment and assets in hospitals. These technologies are also being used in the pharmaceutical industry.

Agriculture. AI and smart algorithms can analyse massive amounts of data on weather, environment and historical information to make increasingly accurate predictions on what, where, when and how to plant crops for optimal performance and yields. The IoT for Agriculture, or IoT4Ag, is a new area of technology development that holds tremendous potential for improving global food production. There is great potential of AI combined with IoT4Ag.

Manufacturing. Industry 4.0, or the fourth industrial revolution, includes integration of the IoT and AI, and a move towards semi-autonomous decisions. AI-powered robots are used in many manufacturing setups.

The IoT and machine learning in predictive maintenance can be found in machines for manufacturing. As per a report from Forbes, 44 per cent of respondents from automotive and manufacturing sectors classified AI as highly important to the manufacturing function in the next five years, while 49 per cent said it was absolutely critical for success.

Wearables. Modern wearable technologies have enhanced sensing and AI-based cognitive abilities, and can create augmented reality.

Wearable sensors enable soldiers to be tracked easily, especially in high-alert operations to monitor their safety.

Military. AI, Big Data and machine

learning are used by the military to safeguard people against rogue entities. These technologies are used to sift through and analyse drone footages and other activities.

AI and machine learning could also be used in cybersecurity to minimise data theft and other malicious activities. AI and the IoT hold the promise for defence, including actionable information and building a complete picture on critical decisions to be made.

Challenges ahead

While AI and the IoT combination has great potential to transform the industry, there are numerous challenges coming to light for implementation and faster

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adoption, including hardware, lack of standardisation and regulations for ethical issues, and interoperability.

For example, an AI-chatbot, which pops up on websites to interact with customers, can either follow a scripted text or use machine learning. If it does not work as per scripted text and has collected sensitive personal information of the customer, or if a surveillance camera exposed personal photos to the outside world, there is no clear provision for accountability. Also, if a self-driving car got involved in an accident, accountability for damage to property or death of a person is unclear.

The IoT sensors are embedded in different devices, including industrial and commercial. It is difficult to synchronise the data flow between all this hardware without the help

of a professional team. It is difficult to calibrate sensors and keep IoT hardware updated on a regular basis. Both software and hardware are equally important in AI and the IoT. Although computing capabilities have advanced significantly in recent years, bottlenecks have shifted to computing hardware such as processors, memories and sensors.

As more and more data is generated and complexity increases, new computing power is required in hardware systems. Newer computer chips and platforms require high-bandwidth memory systems to address the bandwidth needs of AI and related systems. Therefore a number of engineering challenges arise, such as design complexity, high number of inputs and outputs, cost, systems integration and so on.

Latest deep neural network designs have become more accurate, and so their sizes, parameters and number of operations have also increased dramatically. But performance of typical hardware platforms cannot keep pace with new deep neural network designs. Therefore innovations in hardware are required to meet the increasing computational demands in deep neural networks.

The way forward

AI and the IoT are creating unique outcomes and opportunities in Industry 4.0. These technologies are disruptive forces gradually paving their way from health and finance to manufacturing and commerce industry. Use and development of AI and the IoT are not driven by pure academics or research; it is done to address the real needs of the world today.

There is a lot of progress and development being made by various industries to improve these technologies, including memories, processors, sensors, power efficiency devices, machine learning and deep neural networks. However, due to lack of standardisation and regulations, there are certain hurdles that are impacting governments, enterprises, institutions and individuals in multiple ways. To get maximum benefits from these technologies, a set of governance frameworks is needed for a holistic approach. **EFY**